

Lecture Notes:

Introduction, Asset Pricing Guidance and Methodologies

Expanded notes based on slides by Yanzhi Wang
with in-class points from 10 September 2026

These notes follow the original twenty-four slides, add implementation detail, and incorporate points emphasized in class: AR processes and long memory, compounding conventions, market-maker microstructure, Fama–French three-factor model, idiosyncratic risk, Altman zones, return skewness, Type I/II error, look-ahead bias, delisting, lagged weights, SEO effects, and measurement error in beta. American English spelling is used throughout.

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1. Title and purpose

The deck is an introductory map of empirical asset pricing: how to measure returns correctly, how portfolio construction and market microstructure distort those measurements, how CAPM language is used even when the model is rejected, and how to keep statistical results economically interpretable. The closing claim of the whole sequence is that a clean research paper is distinguished from a noisy empirical exercise mainly by timing, implementability, and interpretation.

Before any estimation, fix the *direction of the paper*: time series, cross section, or both. That choice determines the objects you report (alphas and betas versus characteristic-sorted average returns), the benchmark model, and the inference.

2. The central empirical asset-pricing question

Empirical work almost always answers one of two questions, and sometimes both.

- **Cross-section.** Why do some assets or portfolios have higher *average* returns than others? Typical objects: $\mathbb{E}(R_i)$, size, book-to-market, momentum, profitability, and other characteristics.
- **Time series.** Do a set of risk factors explain *co-movement* of returns over time? Typical objects: betas, alphas, factor loadings, and how those loadings change with conditioning variables.

A paper should state which question it is answering. The two questions are not interchangeable. A high average return in the cross section can be a risk premium or mispricing. A time-series alpha can be skill, an omitted factor, or a bad benchmark.

Key distinction.

- Cross section: compare average returns across assets or portfolios.
- Time series: explain return fluctuations across time using factors.
- Joint-test issue: an abnormal return may reflect mispricing *or* a misspecified benchmark model.

That last point is the classic joint-hypothesis problem: market efficiency cannot be rejected without also specifying—and potentially misspecifying—the model of risk.

3. Return computation: simple and log returns

The simple (arithmetic) return measures the percentage wealth change over one period. Total return includes both capital gain and distributions:

$$R_t = \frac{P_t + D_t}{P_{t-1}} - 1. \quad (1)$$

This is the object an investor actually experiences and the object that should be used when a paper reports “the strategy earned 1.2 percent last month.”

The log (continuously compounded) return is

$$r_t = \ln(1 + R_t). \quad (2)$$

Log returns add over time, which is why they are convenient for multi-period aggregation and for some econometric specifications:

$$1 + R_{0,T} = \prod_{t=1}^T (1 + R_t), \quad r_{0,T} = \sum_{t=1}^T r_t. \quad (3)$$

Simple returns are easier to interpret. Log returns are easier to aggregate and sometimes easier to model. They coincide with simple returns only over very short horizons.

Must account for stock splits and dividends. If the raw close is used and splits or cash dividends are ignored, measured returns are wrong, often by tens of percent on event days.

Always use a total-return series or apply the split factor F_t and the dividend explicitly:

$$R_t = \frac{P_t F_t + D_t}{P_{t-1}} - 1.$$

Ask of every dataset: adjusted price, total return, or raw close?

4. Stock splits, dividends, and adjusted returns

A split changes the number of shares and the price per share, but does not create a true gain or loss. A cash dividend transfers value from the firm to the shareholder; the total return must include the dividend.

The standard schematic is as follows. Price starts at 100. A 2-for-1 split moves the observed price to about 50. Without adjustment the measured return looks like -50% . After the split a dividend of 2 is paid and the end price is 53. With both the split factor and the dividend,

$$R_t = \frac{53 + 2}{50} - 1 = 10\%. \quad (4)$$

CRSP and most professional vendors supply total returns; many free download sites do not.

5. Daily, weekly, and monthly compounded returns

Returns do not scale linearly across time; they compound. Multi-day returns must be built from daily *gross* returns,

$$R_{t,t+h} = \prod_{k=1}^h (1 + R_{t+k}) - 1. \quad (5)$$

Trading-day conventions used in class. Equity research almost always compounds over *trading* days, not calendar days:

Horizon	Trading days
1 week	5
1 month	21
1 quarter	63
1 year	252

So a one-month compounded return from daily data is the product of 21 daily gross returns minus one, not “thirty calendar days of simple addition.” A one-year return is the product of 252 daily gross returns minus one. These conventions also govern how volatilities and Sharpe ratios are annualized: $\sigma_{\text{ann}} \approx \sigma_{\text{daily}} \sqrt{252}$.

Sampling frequency affects noise, microstructure bias, and the interpretation of autocorrelation. Daily series are noisier and more contaminated by bid–ask bounce and stale prices. Monthly series are closer to an economic object, which is why most cross-sectional asset-pricing work is conducted at monthly frequency.

6. Autocorrelation, the AR(1) process, and long memory

Campbell, Lo, and MacKinlay (1997), *The Econometrics of Financial Markets*, report first-through fourth-order sample autocorrelations of CRSP value-weighted (VW) and equal-weighted (EW) indexes.

Table 1: Sample autocorrelations of CRSP indexes.

Frequency	Index	$\rho(1)$	$\rho(2)$	$\rho(3)$	$\rho(4)$
Daily	CRSP Value-Weighted	17.6%	-0.7%	0.1%	-0.8%
Daily	CRSP Equal-Weighted	35.0%	9.3%	8.5%	9.9%
Weekly	CRSP Value-Weighted	1.5%	-2.5%	3.5%	-0.7%
Weekly	CRSP Equal-Weighted	20.3%	6.1%	9.1%	4.8%
Monthly	CRSP Value-Weighted	4.3%	-5.3%	-1.3%	-0.4%
Monthly	CRSP Equal-Weighted	17.1%	-3.4%	-3.3%	-1.6%

Two facts stand out. First, autocorrelation generally declines as the horizon lengthens from daily to monthly. Second, as emphasized in class, the AR of the equal-weighted index is stronger than the AR of the value-weighted index at every frequency.

6.1 The AR(1) process

A first-order autoregression is the workhorse short-memory model for a mean-zero (or demeaned) return r_t :

$$r_t = \phi r_{t-1} + \varepsilon_t, \quad |\phi| < 1, \quad \varepsilon_t \sim \text{WN}(0, \sigma_\varepsilon^2). \quad (6)$$

The implied autocorrelations decay *geometrically*:

$$\rho(k) = \phi^k, \quad k = 1, 2, \dots \quad (7)$$

If $\phi = 0.35$ (the daily EW $\rho(1)$ in the table), an AR(1) would predict $\rho(2) = 0.123$, $\rho(3) = 0.043$, $\rho(4) = 0.015$. The actual daily EW coefficients (9.3%, 8.5%, 9.9%) decay much more slowly than that. The VW daily series is closer to a short-lived AR(1) that is essentially gone after one lag.

Stationarity of AR(1) requires $|\phi| < 1$. The unconditional variance is $\text{Var}(r_t) = \sigma_\varepsilon^2 / (1 - \phi^2)$. The half-life of a shock is $\ln(0.5) / \ln |\phi|$.

6.2 Long memory

Short memory means the autocorrelation is absolutely summable, $\sum_{k=1}^{\infty} |\rho(k)| < \infty$, as in any stationary ARMA process. Long memory means the ACF decays *hyperbolically*,

$$\rho(k) \sim C k^{2d-1}, \quad 0 < d < 1/2, \quad (8)$$

so the autocorrelations are not summable. The leading parametric model is fractional integration, $I(d)$, or ARFIMA. Long memory is not the same thing as a unit root ($d = 1$); it is a slow but still mean-reverting dependence.

In this course the practical point is diagnostic. Daily EW index returns look “longer memory” than daily VW returns: $\rho(1)$ is larger *and* $\rho(2)$ – $\rho(4)$ stay positive. That pattern is largely microstructure and lead–lag (below), not a claim that small-stock prices are fractionally integrated. Still, if you fit only an AR(1) to an EW daily series you will understate persistence at lags 2–4. Check the ACF, not just $\rho(1)$.

6.3 Why the equal-weighted index has stronger positive $\rho(1)$

The gap is not primarily “small stocks are more predictable in an economic sense.” It is a market-microstructure and portfolio-construction effect.

Nonsynchronous trading and stale prices. An index return at the close is computed from the last transaction price of each constituent. Large, liquid stocks that dominate the value-weighted index trade almost continuously, so their last price is a timely close. Small stocks that receive equal weight in the EW index trade infrequently. Their last recorded price is often hours—or even a day—old. When news arrives late in the day, large stocks move immediately. Small stocks that have already stopped trading do not. The next morning, or the next day they trade, they catch up. That delayed adjustment produces positive serial correlation in the *portfolio* return even if every individual stock is close to a martingale.

Lo and MacKinlay (1990) and Campbell, Lo, and MacKinlay (1997) formalize the point: the induced first-order autocorrelation of a portfolio is an increasing function of the non-trading probabilities of its constituents. Equal weighting loads far more on high-non-trading names.

Cross-autocorrelations and lead-lag effects. Large stocks lead small stocks. Today’s return on big, frequently traded firms forecasts tomorrow’s return on small, thinly traded firms. Because EW puts equal weight on the lagging small names, this cross-effect appears as own-autocorrelation of the EW index. Lo and MacKinlay (1990) show that a substantial fraction—on the order of half—of the positive index autocorrelation can be attributed to these positive cross-effects rather than to own-autocorrelation of the individual stocks. Individual stocks actually tend to have *negative* first-order autocorrelation because of bid-ask bounce (Section 13).

Portfolio averaging itself. Even if every stock has zero or negative own-autocorrelation, averaging many stocks with heterogeneous trading frequencies and positive contemporaneous correlation can induce positive portfolio autocorrelation. The effect is stronger under equal weights than under value weights, because value weighting concentrates the portfolio on the most liquid names.

Why the effect shrinks from daily to monthly. Over a month almost every stock has traded many times. Stale-price and non-trading biases become small, so $\rho(1)$ falls for both indexes. EW remains higher because small stocks still trade less often than large stocks.

Research implication. Daily EW portfolio returns are not a clean laboratory for testing economic predictability. A large fraction of the 35 percent daily $\rho(1)$ is spurious. Serious cross-sectional work usually uses monthly returns, lagged weights, and—when daily data are unavoidable—Dimson or Scholes–Williams betas that add leads and lags of the market. A strategy that “works” on daily EW sorts often disappears after costs or at monthly frequency.

7. Portfolio returns: equal-weighted and value-weighted

Equal-weighted portfolios assign the same weight to each stock. Value-weighted portfolios assign weights proportional to lagged market equity:

$$R_{\text{VW},t} = \sum_i w_{i,t-1} R_{i,t}, \quad w_{i,t-1} = \frac{\text{ME}_{i,t-1}}{\sum_j \text{ME}_{j,t-1}}, \quad (9)$$

$$R_{EW,t} = \frac{1}{N_t} \sum_{i=1}^{N_t} R_{i,t}. \quad (10)$$

Big warning: use lagged weights, never current weights. Weights must be dated $t - 1$ so they are observable at the formation moment. Using $w_{i,t}$ (current market equity, current characteristic, current membership) is look-ahead. It lets the portfolio “know” this period’s price change before the return is measured, and it mechanically inflates backtested performance. This is one of the most common ways an empirical paper becomes invalid.

Empirically, equal-weighted strategies tilt toward small stocks and usually require more frequent rebalancing. Value-weighted portfolios are closer to investable market benchmarks and involve lower turnover. Neither scheme is “wrong”; they answer different questions. Many anomalies look stronger in EW sorts precisely because of the small-stock tilt and the microstructure effects of the previous section.

8. Rebalancing, turnover, and transaction cost

Rebalancing restores target portfolio weights after returns have changed the composition of the portfolio. The benefit is conceptual clarity and stable exposure. The cost is turnover and implementation friction. A strategy that survives only before costs may not be economically meaningful:

$$R_{p,t}^{\text{net}} = R_{p,t}^{\text{gross}} - \text{TC}_t. \quad (11)$$

What raises trading cost?

- Bid–ask spread
- Market impact
- Commissions and fees
- Short-sale constraints
- Taxes and financing frictions

A paper that reports only gross returns on an equal-weighted, monthly-rebalanced small-stock strategy is describing a paper portfolio, not an implementable one.

9. CAPM: intuition

The CAPM says expected return depends only on systematic risk, not on total volatility:

$$\mathbb{E}(R_i) = R_f + \beta_i [\mathbb{E}(R_M) - R_f], \quad \beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}. \quad (12)$$

Investors are compensated for bearing market risk because idiosyncratic risk can be diversified away. The working time-series regression is

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i (R_{M,t} - R_{f,t}) + \varepsilon_{i,t}. \quad (13)$$

The market portfolio becomes the unique tangency portfolio when all investors hold mean-variance efficient portfolios (two-fund separation). Even papers that reject the CAPM still use

this language— alpha, beta, systematic versus idiosyncratic—as the benchmark vocabulary.

10. Fama–French three-factor model

The CAPM is the language. The workhorse empirical benchmark after 1993 is the Fama–French three-factor model (Fama and French 1993):

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,M}(R_{M,t} - R_{f,t}) + \beta_{i,s} \text{SMB}_t + \beta_{i,h} \text{HML}_t + \varepsilon_{i,t}. \quad (14)$$

- $R_M - R_f$: market excess return.
- SMB (small minus big): return on a diversified portfolio of small stocks minus a diversified portfolio of big stocks. It is the size factor.
- HML (high minus low): return on high book-to-market (value) stocks minus low book-to-market (growth) stocks. It is the value factor.

Interpretation: α_i is now abnormal return *after* size and value exposures, not just after market beta. Many CAPM alphas of small and value portfolios shrink once SMB and HML are included. Later extensions add momentum (Carhart WML/UMD), profitability (RMW), and investment (CMA). In a time-series paper, report the full set of loadings and the Gibbons–Ross–Shanken (GRS) test that all alphas are jointly zero. In a cross-sectional paper, the three factors are the competing risk explanation for characteristic-sorted spreads.

11. Capital Market Line versus Security Market Line

The Capital Market Line uses total risk on the horizontal axis and applies only to efficient portfolios:

$$\mathbb{E}(R_p) = R_f + \frac{\mathbb{E}(R_M) - R_f}{\sigma_M} \sigma_p. \quad (15)$$

The Security Market Line uses beta on the horizontal axis and applies to any asset or portfolio:

$$\mathbb{E}(R_i) = R_f + \beta_i [\mathbb{E}(R_M) - R_f]. \quad (16)$$

Students often confuse them because both are straight lines that pass through R_f and the market point. They are related but conceptually different objects: one describes total risk of efficient combinations; the other describes priced systematic risk of everything.

The empirical SML is also where measurement error in beta shows up (Section 22): noisy betas flatten the estimated SML even if the true CAPM holds.

12. Market microstructure and the market maker

Microstructure studies how trading mechanisms, liquidity suppliers, and order flow affect observed prices and returns. Remember the market maker. The specialist or dealer posts a bid (the price at which the maker *buys*) and an ask (the price at which the maker *sells*). The spread is the maker’s compensation for three things:

- order-processing costs,
- inventory risk (the maker may be forced to hold an unwanted long or short position),
- adverse selection (the incoming order may be informed).

Observed transaction prices are not the latent efficient price. They are the bid or the ask, depending on whether the incoming order was a sell or a buy. Write

$$\text{Spread}_t = \text{Ask}_t - \text{Bid}_t, \quad \text{Midquote}_t = \frac{\text{Ask}_t + \text{Bid}_t}{2}. \quad (17)$$

The midquote is usually a better proxy for efficient value than the last trade. At high frequencies this distinction dominates estimated returns, volatility, and autocorrelation. Even at daily frequency it still matters for small stocks.

13. Bid–ask spreads and bid–ask bounce

If one trade occurs at the ask and the next at the bid, the measured return can be negative even when fundamental value is unchanged.

Stylized example: trade 1 at the ask of 101, trade 2 at the bid of 99. Observed return $= 99/101 - 1 \approx -1.98\%$.

This bid–ask bounce is one reason daily *individual-stock* returns can display negative first-order autocorrelation. Portfolio returns usually show the opposite sign because the nonsynchronous-trading and lead–lag effects of Section 6 dominate the bounce.

Important lesson. Some short-horizon “return predictability” is microstructure, not economics.

14. Portfolio performance measures

Different performance measures divide return by different notions of risk.

Table 2: Performance measures and their risk denominators.

Measure	Numerator	Risk denominator
Sharpe	Excess return	Total volatility
Treynor	Excess return	Beta
Jensen alpha	Abnormal return	No ratio (a level)
Info. ratio	Alpha	Tracking error

For a fully diversified portfolio, Sharpe and Jensen alpha are often the most informative starting points. For *assessing an active portfolio against a benchmark*, the two objects emphasized in class are the information ratio and tracking error.

14.1 Sharpe ratio and Treynor ratio

$$\text{SR}_p = \frac{\mathbb{E}(R_p - R_f)}{\sigma(R_p - R_f)}, \quad \text{TR}_p = \frac{\mathbb{E}(R_p - R_f)}{\beta_p}. \quad (18)$$

Sharpe uses total risk and is the right comparison when the portfolio is the investor’s entire risky holding. Treynor uses only systematic risk.

14.2 Jensen alpha, tracking error, and the information ratio

Jensen alpha measures average abnormal return relative to a benchmark model such as the CAPM or Fama–French:

$$\alpha_p = \mathbb{E}(R_p - R_f) - \beta_p \mathbb{E}(R_M - R_f), \quad t(\alpha) = \frac{\hat{\alpha}}{\text{SE}(\hat{\alpha})}. \quad (19)$$

A positive alpha can indicate skill, mispricing, omitted risk factors, or model misspecification.

Tracking error is the volatility of the residual (active) return:

$$\text{TE}_p = \sigma(\varepsilon_p) = \sigma(R_{p,t} - R_{b,t}), \quad (20)$$

or, in a factor regression, $\sigma(\hat{\varepsilon}_{p,t})$. It is the risk of departing from the benchmark.

The information ratio scales alpha by that tracking-error volatility:

$$\text{IR}_p = \frac{\alpha_p}{\sigma(\varepsilon_p)} = \frac{\alpha_p}{\text{TE}_p}. \quad (21)$$

IR is the natural score for benchmarked active management: how much abnormal return is produced per unit of active risk. High IR with modest alpha can be more valuable than a large but noisy alpha.

15. Raw risk, systematic risk, and idiosyncratic risk

This section is foundational. Idiosyncratic risk must be defined with the math, not only with words.

Definitions.

- **Raw (total) risk:** $\sigma(R_i)$ or $\text{Var}(R_i)$. Everything an undiversified holder of asset i bears.
- **Systematic risk:** the component of return that co-moves with common factors. In the CAPM this is $\beta_i(R_{M,t} - R_{f,t})$. In Fama–French it is the fitted part from $R_M - R_f$, SMB, and HML.
- **Idiosyncratic risk:** residual risk unique to the asset after the factors are taken out. It is diversifiable in a large portfolio under the usual linear factor model.

Single-factor math. Start from

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{M,t} - R_{f,t}) + \varepsilon_{i,t}, \quad \mathbb{E}(\varepsilon_{i,t} \mid R_{M,t}) = 0. \quad (22)$$

Take variances:

$$\text{Var}(R_{i,t}) = \beta_i^2 \text{Var}(R_{M,t}) + \text{Var}(\varepsilon_{i,t}). \quad (23)$$

The first term on the right is systematic variance. The second term,

$$\sigma_{\varepsilon,i}^2 := \text{Var}(\varepsilon_{i,t}), \quad (24)$$

is idiosyncratic variance. Idiosyncratic volatility (IVOL) is $\sigma_{\varepsilon,i}$. In a sample of T observations after a regression with k factors (including the intercept),

$$\hat{\sigma}_{\varepsilon,i}^2 = \frac{1}{T-k} \sum_{t=1}^T \hat{\varepsilon}_{i,t}^2. \quad (25)$$

R^2 split.

$$R_i^2 = \frac{\beta_i^2 \text{Var}(R_M)}{\text{Var}(R_i)} = 1 - \frac{\text{Var}(\varepsilon_i)}{\text{Var}(R_i)}. \quad (26)$$

A low R^2 means most of the stock's variance is idiosyncratic.

Multi-factor math. With Fama–French three factors, write $f_t = (R_{M,t} - R_{f,t}, \text{SMB}_t, \text{HML}_t)'$ and $\beta_i = (\beta_{i,M}, \beta_{i,s}, \beta_{i,h})'$. Then

$$\text{Var}(R_i) = \beta_i' \text{Var}(f_t) \beta_i + \text{Var}(\varepsilon_i). \quad (27)$$

Idiosyncratic risk is still $\text{Var}(\varepsilon_i)$, but the residual is now orthogonal to three factors rather than one. Adding factors reduces measured IVOL if those factors actually explain co-movement.

Why it is priced or not. Under CAPM/APT logic, idiosyncratic risk is not priced because it can be diversified away. Empirically, a large literature studies whether IVOL *is* priced in the cross section (Ang, Hodrick, Xing, and Zhang 2006 and the debate that followed). For this course, first get the decomposition right: idiosyncratic risk is the residual variance after the chosen factor model, not “any volatility that looks stock-specific.”

16. Default risk: three common approaches

- **Structural approach.** Merton distance-to-default uses equity value, volatility, and the debt threshold. It is market-based and forward-looking.
- **Accounting-score approach.** Altman Z -score predicts distress from accounting ratios. It is a discriminant-score approach.
- **Probabilistic-score approach.** Ohlson O -score estimates default likelihood using firm characteristics. It is a logit-style distress index.

They are complements, not substitutes. Distance-to-default uses market prices and volatility. Z and O use accounting statements that can be stale or managed.

17. Merton model and distance to default

The firm defaults at horizon T if asset value falls below the promised debt payment. Distance to default measures how many asset-volatility units the firm is away from the default boundary:

$$\text{DD} = \frac{\ln(V_A/D) + (\mu_A - \frac{1}{2}\sigma_A^2)T}{\sigma_A\sqrt{T}}. \quad (28)$$

A larger DD implies a lower risk-neutral or physical probability of default, depending on implementation. V_A and σ_A are not observed; they are backed out from equity value and equity

volatility (together with a debt face value) by solving the Merton equations. That inversion is the operational content of Merton DD.

18. Altman Z-score: formula and zones of discrimination

Altman Z-score is a linear combination of accounting ratios, originally designed to classify firms into bankruptcy zones (Altman 1968). The classic manufacturing specification is

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5, \quad (29)$$

where

- X_1 = working capital / total assets,
- X_2 = retained earnings / total assets,
- X_3 = EBIT / total assets,
- X_4 = market value of equity / book value of total liabilities,
- X_5 = sales / total assets.

Zones of discrimination (must know). These cutoffs are about bankruptcy prediction, not about “cheap versus expensive” stocks.

Zone	Z range	Meaning
Safe (non-bankrupt)	$Z > 2.99$	Low probability of failure
Grey	$1.81 < Z < 2.99$	Indeterminate; needs judgment
Distress (bankrupt)	$Z < 1.81$	High probability of failure

A firm with $Z = 3.4$ is in the safe zone. A firm with $Z = 1.2$ is in the distress zone. A firm with $Z = 2.2$ is in the grey zone: the discriminant rule does not classify it cleanly.

O -score is an accounting-based probability index; it is more flexible and often used in large-sample empirical work:

$$O\text{-score} = \beta_0 + \sum_k \beta_k X_k. \quad (30)$$

Neither measure is a structural valuation model. Pedagogical point: DD is market-based; Z-score and O -score are accounting-based.

19. Skewness of stock returns

Skewness is the standardized third central moment:

$$\text{Skew}(X) = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}. \quad (31)$$

Right skew versus left skew (must know the definition).

- **Skewed to the right** (positive skew, $\text{Skew} > 0$): the right tail is longer. Mean $>$ median. A few very large positive outcomes pull the mean up. The mass of the distribution sits to the left of the mean.
- **Skewed to the left** (negative skew, $\text{Skew} < 0$): the left tail is longer. Mean $<$ median. A few very large negative outcomes pull the mean down.

What stock returns do. Simple (arithmetic) stock returns tend to be *skewed to the right*. Limited liability floors a simple return at -100% , while the upside is unbounded. Individual stocks, especially small and lottery-like names, show pronounced positive skewness. Indexes are less right-skewed than single names because the right-tail events are not perfectly correlated, and at daily frequency an index can even look mildly left-skewed because crashes are more synchronized than booms.

Log returns, $r = \ln(1 + R)$, compress the right tail and stretch the left tail, so they are closer to symmetric and often slightly negatively skewed. When a paper says “returns are right-skewed,” check whether it means simple returns or log returns.

Right skew also matters for portfolio evaluation: a high mean can be produced by a handful of extreme winners. Report median returns or higher-moment measures if the strategy is a right-skew lottery.

20. Value-at-Risk and tail risk

Value-at-Risk answers a quantile question: how bad can losses get at a given confidence level?

$$P(L > \text{VaR}_\alpha) = 1 - \alpha, \quad \text{VaR}_\alpha \approx \mu_P + z_\alpha \sigma_P. \quad (32)$$

The normal approximation is only a starting point; empirical and extreme-value versions are used in practice. Because returns are skewed and fat-tailed, normal VaR understates crash risk.

Expected Shortfall answers a severity question: once we are in the tail, how large is the average loss?

$$\text{ES}_\alpha = \mathbb{E}[L \mid L > \text{VaR}_\alpha]. \quad (33)$$

VaR is intuitive. Expected Shortfall often conveys tail risk more completely and is a coherent risk measure (it satisfies subadditivity; VaR does not).

21. Statistical significance, economic significance, and error types

A coefficient can be statistically significant but too small to matter after costs, funding constraints, and implementation frictions. Conversely, a large economic effect may be imprecisely estimated in a small sample. Good empirical asset-pricing research reports both t -statistics and economic magnitudes.

	Low economic significance	High economic significance
Low statistical significance	Ignore or collect more data	Economically large, but imprecisely estimated
High statistical significance	Noisy or small effect	Publishable only if robust and implementable

21.1 Type I error and Type II error

Let H_0 be the null, for example “alpha is zero” or “the anomaly premium is zero.”

	H_0 true	H_0 false
Reject H_0	Type I error (false positive)	Correct rejection (power)
Do not reject H_0	Correct non-rejection	Type II error (false negative)

- **Type I error.** Rejecting a true null. In asset pricing: publishing an “anomaly” that is sampling noise or data snooping. The size of a test is $P(\text{Type I}) = \alpha$, conventionally 5 percent.
- **Type II error.** Failing to reject a false null. In asset pricing: missing a genuine premium because the sample is short, the strategy is noisy, or the test is underpowered. Power is $1 - P(\text{Type II})$.

Multiple testing across many characteristics inflates the effective Type I rate. That is why robustness, out-of-sample periods, and economic magnitude after costs matter as much as a single t -statistic.

22. Measurement error in beta and the errors-in-variables problem

Beta is not observed. It is estimated from a time-series regression, so the regressor in a second-stage cross-sectional regression is measured with error. The two whiteboards from 10 September 2026 are transcribed first, then unpacked.

Left board (SML test and measurement error).

- Sorting variable; formation period; SML sketch from R_f to R_M .
- Fiscal year end of month (timing of accounting sorts).
- Cross-sectional SML: $\mathbb{E}(R_i) - R_f = \gamma_0 + \gamma_1\beta_i + u_i$.
- CAPM restrictions: $\gamma_0 \doteq 0$, $\gamma_1 > 0$ (slope = $\mathbb{E}(R_M) - R_f$).
- Estimated beta equals true slope plus noise:

$$\hat{\beta}_i = \beta + \frac{\sum_t (R_{i,t} - \bar{R}_i) u_{i,t}}{\sum_t (R_{M,t})^2}.$$

The first β is the *true parameter of slope*. The fraction is measurement error V_i .

- Population orthogonality: $\mathbb{E}(\sum (R_i - \bar{R}_i) u_i / \sum R_M^2) = 0$.

Right board (portfolio grouping).

- Step 1: sort into 20 (or 10) portfolios.
- $\hat{\beta}_i = \beta + V_i$, $\hat{\beta}_p = \beta_p + V_p$, $V_p = (1/N_p) \sum_i V_i$.
- Sample model: $y = \alpha + \beta x + \varepsilon$.
- Population: $y = a + bx + u$.

22.1 The cross-sectional SML on the board

After betas have been estimated in a formation (or estimation) period, the CAPM is tested as a cross-sectional Security Market Line

$$\mathbb{E}(R_i) - R_f = \gamma_0 + \gamma_1\beta_i + u_i. \tag{34}$$

Under the CAPM the intercept should be zero and the slope should equal the market risk premium:

$$\gamma_0 \doteq 0, \quad \gamma_1 = \mathbb{E}(R_M) - R_f > 0. \quad (35)$$

If you plot average excess return against beta you should get a line from R_f up through the market point R_M . That is the sketch on the far left of the board. Empirically the fitted line is often too flat; measurement error in β_i is one reason.

The phrase *sorting variable* means the characteristic used to form portfolios before this regression (beta itself, size, book-to-market). *Formation period* is the window in which that sort is done. *Fiscal year end of month* is the timing rule for accounting sorts: book equity and other fiscal-year items are known only after fiscal year-end (in the usual Fama–French calendar, portfolios are formed at the end of June using the previous December statements). Using year-end book equity on 1 January is look-ahead.

22.2 Where the measurement error V_i comes from

Write the time-series market model as

$$R_{i,t} - \bar{R}_i = \beta (R_{M,t} - \bar{R}_M) + u_{i,t}. \quad (36)$$

The OLS slope that you actually compute is

$$\hat{\beta}_i = \frac{\sum_t (R_{i,t} - \bar{R}_i)(R_{M,t} - \bar{R}_M)}{\sum_t (R_{M,t} - \bar{R}_M)^2}. \quad (37)$$

Substitute the market-model expression for $R_{i,t} - \bar{R}_i$. The first piece recovers the true slope β ; the leftover piece is the circled formula on the board,

$$\hat{\beta}_i = \beta + \frac{\sum_t (R_{i,t} - \bar{R}_i) u_{i,t}}{\sum_t (R_{M,t} - \bar{R}_M)^2}. \quad (38)$$

The first β was labeled *true parameter of slope*. The fraction is the measurement error

$$V_i := \frac{\sum_t (R_{i,t} - \bar{R}_i) u_{i,t}}{\sum_t (R_{M,t} - \bar{R}_M)^2}, \quad \hat{\beta}_i = \beta + V_i. \quad (39)$$

In the population the residual is orthogonal to the market, so

$$\mathbb{E} \left(\frac{\sum_t (R_{i,t} - \bar{R}_i) u_{i,t}}{\sum_t R_{M,t}^2} \right) = 0. \quad (40)$$

That is the line under “measurement error V_i .” Orthogonality holds *in expectation*. In any finite sample the fraction is not exactly zero. That nonzero finite-sample term is V_i . The beta you plug into the SML is therefore the true beta plus noise.

22.3 Why that noise flattens the SML

The second-pass regression uses $\hat{\beta}_i = \beta_i + V_i$ as if it were β_i . This is the classical errors-in-variables problem. If V_i is classical measurement error,

$$\text{plim } \hat{\gamma}_1 = \gamma_1 \cdot \frac{\text{Var}(\beta)}{\text{Var}(\beta) + \text{Var}(V)} < \gamma_1. \quad (41)$$

The estimated price of beta risk is attenuated toward zero. A true SML with slope $\mathbb{E}(R_M) - R_f$ looks too flat, and a test that “the CAPM fails because $\hat{\gamma}_1$ is too small” can be an artifact of noisy betas. The problem is worse for individual stocks than for portfolios, because stock-level $\text{Var}(V)$ is large.

The board also wrote the usual regression contrast:

$$\begin{aligned}\text{sample model: } y &= \alpha + \beta x + \varepsilon, \\ \text{population: } y &= a + bx + u.\end{aligned}$$

(The board labeled the second line as the population equation.) β in the sample equation is an estimator of the population slope b . V_i is the reason those two objects are not the same.

22.4 Why “sort into 20 (10) portfolios”

Step 1 on the right board is the standard fix. Sort stocks into 10 or 20 portfolios (by pre-ranking beta, or by size then beta). The portfolio beta is

$$\hat{\beta}_p = \beta_p + V_p, \quad V_p = \frac{1}{N_p} \sum_{i \in p} V_i. \quad (42)$$

If the individual measurement errors are imperfectly correlated, V_p is much smaller than V_i . Portfolio betas are more precise, so the attenuation factor $\text{Var}(\beta)/(\text{Var}(\beta) + \text{Var}(V))$ moves closer to one and the SML slope is less biased. That is why Black–Jensen–Scholes and Fama–MacBeth run the second pass on portfolios, not on thousands of individual stocks.

Other standard repairs, used together with sorting:

- Dimson (1979) or Scholes–Williams betas (leads and lags of the market), which shrink the stale-price part of V_i ;
- an independent second beta estimate as an instrument;
- EIV-robust or Bayesian shrinkage estimators.

Measurement error in beta is not a side issue. It changes whether you conclude that the CAPM “fails” or that the test is biased.

23. The most common empirical asset-pricing mistakes

- Look-ahead bias: using information that was not available at the portfolio formation date.
- Survivorship bias: excluding delisted or failed firms and thereby overstating performance.
- Ignoring delisting returns, stock splits, or dividend adjustments.
- Using contemporaneous portfolio weights instead of observable lagged weights.
- Forgetting transaction costs or short-sale constraints when interpreting anomaly profits.
- Treating statistical significance as sufficient evidence without robustness or economic interpretation.

23.1 Look-ahead bias, in detail

Look-ahead bias means the backtest uses a number that a real investor could not have known on the formation date.

Concrete examples:

- Forming a book-to-market portfolio on 1 January using fiscal-year book equity that is not released until March.
- Using this month's market capitalization, or this month's return, as the weight for this month's portfolio return.
- Using Compustat data that have been restated after the fact.
- Requiring that a firm survive to the end of the sample before it is allowed into the beginning of the sample.
- Using a characteristic that is constructed with future prices (for example a "yield" that embeds the next dividend before the ex-date is known).

The test of look-ahead is operational: write down the exact calendar date on which each input becomes public, and form the portfolio only from inputs dated on or before that date. If the paper cannot produce that calendar, the result is not implementable.

23.2 Delisted firms: do not study only listed survivors

Whenever you do the research, do not just include listed firms. You must include delisted firms. Dropping names that die, merge, or are taken private is survivorship bias. Failed firms have low average returns. A sample of firms that are still alive at the end of the window overstates average returns, understates default risk, and can manufacture a "profitable" strategy that never had to hold the losers through delisting.

In CRSP, use the delisting return (`dlret`) when a name leaves the file. If the delisting return is missing, a conservative convention in many papers is to assign -30% for performance-related delistings (Shumway 1997) and to check robustness. The point is not the exact fill-in rule. The point is that the dead names stay in the sample through the delisting month.

24. SEO effects on the financial market

A seasoned equity offering (SEO) is an issue of new shares by a firm that is already public, as opposed to an IPO. The standard empirical facts, and the reasons they matter for asset pricing, are as follows.

Announcement effect. The stock price typically falls on the SEO announcement day. The usual interpretations are:

- Myers–Majluf adverse selection: managers issue when they believe the stock is overvalued, so the market infers bad news.
- Dilution and capital-structure change.
- Downward-sloping demand for the firm's shares (price pressure).

Long-run performance. SEO firms, on average, underperform matching non-issuers over the subsequent three to five years (Loughran and Ritter 1995 and the large literature that followed). Whether this is mispricing (market timing by managers) or compensation for risk (issuers have different loadings on profitability, investment, or other factors) is the live research question. In

Fama–French language, issuers tend to be high-investment, low-profitability growth firms, so some of the underperformance is absorbed by later factor models.

Why this belongs in an empirical-methods course. SEO studies are a clean laboratory for every warning in these notes. Event windows must not leak future information (look-ahead). The sample must keep firms that later delist. Benchmark returns must use lagged weights. Abnormal performance can be a time-series alpha or a cross-sectional issuer-versus-non-issuer spread; the paper must say which. Announcement-day moves are also a microstructure object: spreads widen and market makers manage inventory around the issue.

25. How to read an empirical asset-pricing paper

1. Start from the question: is the paper about time-series factor pricing, cross-sectional return prediction, or both?
2. Identify the variable of interest, the test assets, the benchmark model, and the economic mechanism.
3. Inspect implementation: sample construction, timing, weighting scheme, rebalancing, delisting treatment, and inference.
4. Ask whether the effect is robust, economically meaningful after costs, and plausibly interpretable—and whether beta or other regressors are measured with error.

Step 3 is where most papers that look impressive on first reading fall apart.

26. Take-aways

- Accurate return measurement is the first step: splits, dividends, compounding over 5 / 21 / 63 / 252 trading days, and delisting returns.
- AR(1) is short memory with geometric ACF decay. Daily EW returns look more persistent than daily VW returns; much of that gap is microstructure and lead-lag, not a tradable long-memory profit.
- Use lagged weights. Never current weights.
- Include delisted firms. A listed-only sample is survivorship bias.
- Remember the market maker: observed prices are bids and asks, not the latent efficient price.
- For active portfolios, report information ratio and tracking error, not only raw return.
- CAPM is the language; Fama–French three-factor is the workhorse benchmark. Idiosyncratic risk is residual variance after the chosen factors.
- Altman Z is a bankruptcy classifier: $Z > 2.99$ safe, $1.81 < Z < 2.99$ grey, $Z < 1.81$ distress.
- Simple stock returns tend to be skewed to the right. Know the definition of right versus left skew.
- Type I error is a false discovery; Type II error is a missed true effect. Both matter.
- Look-ahead bias is using any input that was not public on the formation date.

- Measurement error in beta attenuates the estimated price of risk (errors-in-variables). That is one reason the empirical SML is flat.
- SEO announcement drops and post-issue underperformance are a standard application of all of the above.
- State whether the paper is time series, cross section, or both. Careful timing, implementability, and interpretation are what separate a clean PhD paper from a noisy empirical exercise.

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